Mikhail Maslov, PhD

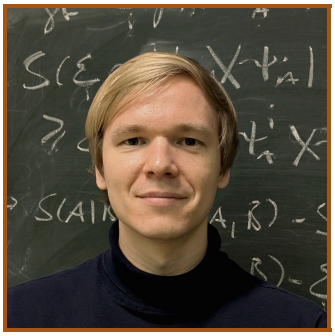
Position: Postdoctoral researcher at
Institute of Science and Technology Austria (ISTA)

Contact: mkmaslov@gmail.com · +43 677 616 59451

Links: Website · Google Scholar · GitHub · LinkedIn

Location: Vienna, Austria

Mobility: Austrian citizen · Open to relocation across Europe



Profile

Theoretical physicist with experience in mathematical modelling, scientific computing and machine learning, seeking quantitative research roles focused on systematic strategies, portfolio construction and risk modelling.

Education

PhD in Theoretical Physics **(2018–2024)**
Institute of Science and Technology Austria (ISTA)
Thesis: *"Emergent physics of rotating quantum impurities in many-body environments"*
Committee: Mikhail Lemeshko (advisor), Benjamin A. Stickler, Maksym Serbyn
Research focus: group theory, mathematical physics, numerical modelling, and PINNs.

BSc in Applied Mathematics and Physics **(2014–2018)**
Moscow Institute of Physics and Technology (MIPT)
Thesis: *"Theory of plasmonic instabilities in MOSFETs with an asymmetric grating-gate"*
Advisor: Dmitry Svintsov
Graduated with Honors (GPA 8.6/10), received Abramov and State Academic fellowships.

Experience

Postdoctoral Researcher **(2025–present)**
Lemeshko group at Institute of Science and Technology Austria (ISTA)
2 publications (+2 more under review); reviewer for *Phys. Rev. A* and *Phys. Rev. Research*

- Built from the ground up a neural-network solver for quantum few-body problems:
  - Implemented Markov chain Monte Carlo (MCMC) sampling and designed network representation that embeds rotational symmetry as a universal inductive bias.
  - Trained the network using system energy as a single objective, avoiding relying on manually tuned or arbitrarily defined hyperparameters.
  - Validated convergence against a DVR benchmark across a broad range of system scales.
- Developed a symmetry-aware mathematical model for optical sensing of material chirality.

PhD Researcher **(2018–2024)**
Lemeshko group at Institute of Science and Technology Austria (ISTA)
4 publications

- Developed a universal group-theoretical framework for analyzing angular momentum exchange in optical systems, applicable to many research fields in physics.
- Formulated analytical models for the experimental control of quantum many-body systems; assessed their feasibility numerically using variational and spectral methods.

Undergraduate Researcher **(2016–2018)**
Svintsov group at Moscow Institute of Physics and Technology (MIPT)

- Developed a theoretical framework based on coupled partial differential equations to analyze plasmonic instabilities in nanostructured semiconductor systems.
- Designed and maintained a MATLAB codebase for electrodynamics simulations of two-dimensional electron systems via spectral methods and Fourier analysis.

Intern **(Jun–Aug 2017)**
Krasheninnikov group at Helmholtz Zentrum Dresden-Rossendorf (HZDR)

- Designed and optimized simulations on an HPC cluster (resulting in a publication):
  - Set up and evaluated molecular dynamics simulations in LAMMPS (via OpenMPI).
  - Performed scattering calculations in SRIM (based on Monte Carlo methods).

Intern **(Jun–Aug 2016)**
Vartaniants group at Deutsches Elektronen-Synchrotron (DESY), Hamburg

- Developed MATLAB code for simulating crystallographic data using Monte Carlo methods and comparing it to the experimental data from free-electron laser via Fourier analysis.

Student Representative and Teaching Assistant **(2019–2022)**
Institute of Science and Technology Austria (ISTA)

- Designed curriculum and taught atomic and condensed matter physics to 50+ PhD students.
- Managed curriculum and student infrastructure development (500+ PhD students).
- Organized networking and scientific outreach events.

Teaching assistant **(2014–2019)**
Moscow Institute of Physics and Technology (MIPT)

- Taught mathematics, physics and computer science to 100+ high-school students.

Skills

Mathematics and modelling:
linear algebra, tensor calculus, probability theory, differential equations, mathematical physics, group theory, differential geometry, quantum algorithms, many-body spin systems.

Numerical methods and ML:
Monte Carlo methods, variational methods, finite-difference methods, Fourier analysis, optimization, supervised and reinforcement learning.

Programming: Python (NumPy, SciPy, PyTorch), MATLAB and Wolfram Mathematica

Computing: vectorization, parallel computing, HPC, Linux, Docker, Bash automation.

Interests: game theory, decision theory, auction theory; market making and risk management.

Publications

Publications: 7 publications (+2 under review); Citations: 170 (from Google Scholar)

1. M. Hrast, G. M. Koutentakis, M. Maslov, and M. Lemeshko
   Bottom-up Analysis of Rovibrational Helical Dichroism
   Physical Review Letters **136**, 053204 (2026)

2. M. Bahl, G. M. Koutentakis, M. Maslov, T. Jungnickel, T. Gaßen, M. Lemeshko, and O. H. Heckl
The R-index: a universal metric for evaluating OAM content and mode purity in optical fields
Journal of Physics: Photonics **8**(1), 015071 (2026)
3. A. S. Kumar, M. Maslov, M. Lemeshko, A. G. Volosniev, and Z. Alpichshev
Massive Dirac-Pauli physics in lead-halide perovskites
npj Quantum Materials **10**, 37 (2025)
4. M. Maslov, G. M. Koutentakis, M. Hrast, O. H. Heckl, and M. Lemeshko
Theory of angular momentum transfer from light to molecules
Physical Review Research **6**(3), 033277 (2024)
5. M. Maslov, M. Lemeshko, and A. G. Volosniev
Impurity with a resonance in the vicinity of the Fermi energy
Physical Review Research **4**(1), 013160 (2022)
6. M. Maslov, M. Lemeshko, and E. Yakaboylu
Synthetic spin-orbit coupling mediated by a bosonic environment
Physical Review B **101**(18), 184104 (2020)
7. S. Kretschmer, M. Maslov, S. Ghaderzadeh, M. Ghorbani-Asl, G. Hlawacek, and A. V. Krasheninnikov
Supported Two-Dimensional Materials under Ion Irradiation: The Substrate Governs Defect Production
ACS Applied Materials and Interfaces **10**(36), 30827 (2018)

Conferences

1. **Quantum Metrology in Many-body and Open Systems (QuMIMOS 2026)**
Poster · February 9–13, 2026 · Vienna, Austria
2. **Scientific visit to University of Lisbon (group of M. Silveirinha)**
Talk · November 27, 2025 · Lisbon, Portugal
3. **QuantA Cluster of Excellence (Workshop #5)**
Poster · November 5–7, 2025 · Vienna, Austria
4. **VCQ and quantA Summer School: Many Body Quantum Physics**
Poster · September 15–19, 2025 · Vienna, Austria
5. **Joint Annual Meeting of ÖPG and SPS 2025**
Talk · August 18–22, 2025 · Vienna, Austria
6. **Dynamics of Electrons in Atomic and Molecular Nanoclusters (2025)**
Talk · May 21–27, 2025 · Erice, Italy
7. **Scientific visit to Humboldt University (group of A. Rauschenbeutel)**
Talk · March 6, 2025 · Berlin, Germany
8. **Quantum Physics with Trapped Particles**
Poster · October 13–18, 2024 · Ascona, Switzerland

9. **Scientific visit to University of Heidelberg (group of S. Jochim)**
   Talk · September 19, 2024 · Heidelberg, Germany
10. **Scientific visit to Max Born Institute (group of O. Smirnova)**
    Talk · September 17, 2024 · Berlin, Germany
11. **Cold and Controlled Molecules and Ions 2024 (CCMI 2024)**
    Poster · September 2–6, 2024 · Vienna, Austria
12. **QuantA Cluster of Excellence (Workshop #2)**
    Poster · June 13–14, 2024 · Vienna, Austria
13. **VCQ Summer School 2023: Hybrid Quantum Systems**
    Poster · August 28–September 1, 2023 · Vienna, Austria
14. **Scientific visit to Harvard University (groups of J. Doyle, K.-K. Ni, and M. Lukin)**
    Poster · August 14–16, 2023 · Cambridge, MA, United States
15. **Gordon Research Conference: Quantum Control of Light and Matter**
    Poster · August 6–11, 2023 · Newport, RI, United States
16. **VCQ Colloquium (Student Talk): Ro-vibrational spectroscopy of molecules using vortex beams of light**
    Talk · April 24, 2023 · Vienna, Austria
17. **Minerva-Gentner Symposium on Coherent manipulation of few-body complexes**
    Poster · February 20–23, 2023 · Rehovot, Israel
18. **1st CAPS School on Ultracold Atoms**
    Poster · November 16–18, 2022 · Barcelona, Spain
19. **EMMI Workshop on Long-Range Ultracold Interactions 2022**
    Poster · September 6–9, 2022 · Innsbruck, Austria
20. **VCQ Summer School: Concepts and Applications of Quantum Information**
    Poster · August 28–September 2, 2022 · Vienna, Austria
21. **Quantum Fluids of Light and Matter 2022 (QFLM 2022)**
    Poster · July 2–7, 2022 · Varenna, Italy
22. **Fermi Polarons: from ultracold gases to 2D semiconductors**
    Poster (online) · February 7–18, 2022 · Madrid, Spain
23. **VCQ Colloquium (Student Talk): An impurity with a resonance in the vicinity of the Fermi energy**
    Talk · November 29, 2021 · Vienna, Austria
24. **VCQ Summer School: Quantum Sensing and Imaging**
    Poster · September 6–10, 2021 · Vienna, Austria
25. **Quantum 2021: Dynamics and Local Control of Impurities in Complex Quantum Environments**
    Poster (online) · August 23–September 17, 2021 · Paris, France

26. **VCQ Summer School: Gravitational Quantum Physics**
    Poster (online) · September 3–6, 2020 · Vienna, Austria
27. **Munich Conference on Quantum Science and Technology (MCQST 2020)**
    Poster (online) · July 6–8, 2020 · Munich, Germany
28. **SFB BeyondC: Austrian Quantum Information Conference 2019**
    Poster · October 31, 2019 · Vienna, Austria
29. **Vienna Graduate Conference on Complex Quantum Systems 2019**
    Poster · October 28–30, 2019 · Vienna, Austria
30. **Vienna Summer School on Complex Quantum Systems 2019**
    Poster · September 16–20, 2019 · Vienna, Austria
31. **Introductory Course on Ultracold Quantum Gases 2019**
    Participation · July 8–10, 2019 · Innsbruck, Austria
32. **DPG Spring Meeting of the Condensed Matter Section (SKM)**
    Poster · March 31–April 5, 2019 · Regensburg, Germany
33. **Doctoral Summer School: Nanophotonics and Metamaterials**
    Poster · May 28–June 1, 2018 · St. Petersburg, Russia
34. **60th Scientific Conference at MIPT**
    Talk · November 20–25, 2017 · Moscow, Russia
35. **First Russian Crystallographic Congress**
    Poster · November 21–26, 2016 · Moscow, Russia